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05/10:-

Terraform provider file:-

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}
provider "aws" {
  region = "us-east-1"
}
```

Main Commands:-

→terraform init
→terraform validate
→terraform plan
→terraform apply
→terraform destroy.

To create ec2:-

```
resource "aws_instance" "demo" {
  ami = "ami-id" ⇒ Take from ami catalog
  instance_type = "t3.micro"
  tags = {
    purpose = "trial instance"
  }
}
```

To create key pairs → Search key pairs in console.

For default yes :- --auto-approve

Create VPC

aws_vpc (on documentation)

vpc.tf

```
resource "aws_vpc" "demovpc" {
    cidr_block = "13.0.0.0/16"
    instance_tenancy = "default"

    tags = {
        Name = "learnvpc"
    }
}
```

Subnet

aws_subnet

```
resource "aws_subnet" "pri-sub" {
    vpc_id = aws_vpc.demovpc.id
    cidr_block = "13.0.1.0/24"

    tags {
      Name = "private subnet"
    }
}
```

(bcs vpc id will be created only after vpc is created. So here we use reference)

Copy paste again for 2nd subnet

```
resource .... "pub-sub"
    :
    cidr    = "13.0.2.0/24"
    :
```

__/__/__

Route Table

aws_route_table

Cannot be created before Internet Gateway

Whatever subnet is attached to IGW is the public

aws_internet_gateway

```
resource "aws_internet_gateway" "igw" {
    vpc_id = aws_vpc.demovpc.id

    tags={
      Name = "learn-IGW"
    }
}
```

aws_route_table

```
resource "aws_route_table" "route-table" {
    vpc_id = aws_vpc.demovpc.id

    route = {
      cidr_block = "0.0.0.0/0"
      gateway_id = aws_internet_gateway.igw.id
    }

    tags = {
      Name = "learn-RT"
    }
}
```

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aws_route_table_association

resource "aws_route_table_association" "rt-associate" {
  subnet_id = aws_subnet.pub-sub.id
  route_table_id = aws_route_table.route_table.id
}

ec2.tf
↳
resource {
  :
  subnet_id = aws_subnet.pub-sub.id

ec2.tf
aws_security_group

resource "aws_security_group" "allow_tls" {
  :
  tags = {
    Name = "learn-SG"
  }
}

resource "aws_vpc_security_group_ingress_rule" "allow_https_IPv4" {
  :
  from_port = 443
  to_port = 443
}

(1)

(2) port 22 (tcp)

(3) port 80 (http)